Training chair with double-sided tables

M11 Group

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Introduction

Our designed product is "***Training chair with double-sided table***", aiming at offering users integrated rest-and-office space. The highlight is the double-sided table, which ensures a more robust, and user-friendly structure than the single-sided one already on the market.

In consideration of user requirements and corresponding inputs, outputs are figured out and proper materials are selected. Finally, verification method is proposed.
The design and manufacture of components and devices are regulated under **international standards** for quality management.

Overview

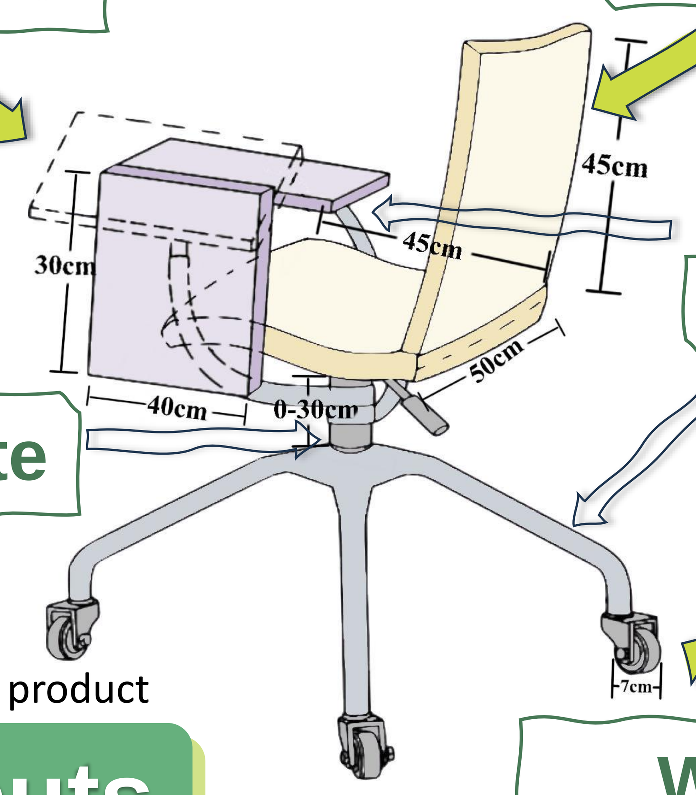

Figure 1. The overview of the product

User Requirements & Inputs & Outputs

Table 1. Ranking & user requirements & corresponding inputs

Ranking	User Requirements	Design Inputs
1	Comfortable resting place	Load-bearing support
		Shape of seat & back rest
		High resilience materials
2	Save space	Table and chair link
3	User-friendly tabletop	Large desk area
		Frictional surface material
		Robust support structure
4	Low potential risks	Air rod type
		High strength
		Non-toxic material
5	Height adjustment	Height-adjusted air rod
6	Removable	Wheels on the bottom
7	Environmentally friendly	Recyclable materials
8	Low cost	Cheap materials
9	Serviceability	Split design
10	Reliability & Durability	Well fatigue-resistant materials
		Robust construction
11	Simple of use	Connect method of two desk

***Possible risks (avoided):** Air rod explosion, Seat collapses or sinks

Note: The **number** of the output corresponds to **ranking** for user requirement

- **Seat&Backrest** (PP/PS/Wood)
  - 45*50cm ⊖ 1
  - "C" Curve ⊖ 1
  - Load capacity >150kg ⊖ 4
  - Fatigue limits>10000 ⊖ 10
- **Double-sided Table** (Stainless Steels/ Carbon Fibre/Brass)
  - 40*30 cm Double-table ⊖ 3
  - Load capacity>75kg ⊖ 4
  - Friction coefficient 0.3-0.6 ⊖ 3&11
  - Snap joint connection ⊖ 3&11
- **Air Rod**
  - Adjustment 0-30cm ⊖ 5&11
  - Load capacity >180kg ⊖ 4&5
  - Baseplate ⊖ 4&5
- **Wheel** (PA/POM/PTFE)
  - Diameter ~7cm ⊖ 6&9
  - Friction coefficient 0.4-0.7 ⊖ 6&11
  - Fixed ⊖ 6&11
  - Load capacity >50kg ⊖ 4&6
- **Bracket** (Aluminium/Titanium Alloy/ Magnesium alloy)
  - Quadrangular structure ⊖ Load capacity >180kg ⊖ 4
  - Hollow structure diameter ~2cm ⊖ 4
  - Rotatable armrests ⊖ Load capacity>~80kg ⊖ 2&4

The materials selected are in accordance with ***ISO 18604：2013. 7***

Components: Removable & Reassembled **2&9**

Materials Selection & Manufacture

Seat & Table & Backrest

Materials Selection

Polycarbonate → Expensive
Wood → Easy-damaged

Polypropylene ~ Best

Manufacturing

Compression Process → Cooling → Demoulding

Bracket

Materials Selection

Titanium Alloy → Difficult to process
Magnesium Alloy → Lower strength

Aluminum Alloy ~ Best

Manufacturing[2]

Melting → Casting → Extrusion → Stretching → Cutting

Air rod & Base plate

Materials Selection

Carbon Fiber → Expensive
Brass → Relatively heavy

Stainless Steel ~Best

Manufacturing

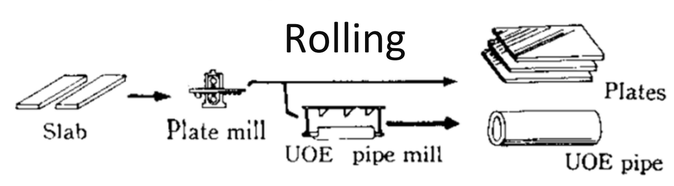

Figure 2. Manufacturing process and properties of stainless steels.[1]

Wheels

Materials Selection

PTFE → Low friction coefficient
Polyoxymethylene → Lower toughness

Polyamide ~ Best

Manufacturing

Melting & Mixing → Injection molding → Shaping → Surface treatment

Verification Plan

✓ **Strength & Stability test**

Static load testing: Applying sustained static force or load to the product.

→ **Yield stress**
Polypropylene > 30 MPa
Aluminum Ally > 300MPa
Stainless Steel > 200MPa
Polyamide > 50MPa

✓ **Friction factor test**

Dynamic friction tester: Measure the required force & angle to calculate the coefficient of dynamic friction[3].

→ **Friction coefficient**
Polypropylene：0.3-0.6
Polyamide：0.4-0.7

✓ **Fatigue test**

Single-point fatigue test method: Put the sample into the bending fatigue testing

→ **Fatigue limit**
Polypropylene > 10000

Reference

[1] Ono Y, Kaito H. Manufacturing process and properties of stainless steels [Internet]. 1986 [cited 2023 Dec 11]. Available from: https://www.jfe-steel.co.jp/archives/en/ksc_giho/no.14/e14-001-011.pdf
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[3] Vanlanduit S, Vanherzeele J, Longo R, Guillaume P. A digital image correlation method for fatigue test experiments. Optics and Lasers in Engineering. 2009;47(3):371-8.

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